

B2C Backend Plan

Separate Laravel shop API · Zoho Payments · B2B fulfilment

Audience	Engineering, product, and operations
Scope	New B2C backend — own server and database
Related system	Existing B2B Laravel app (Mazing Business)
Frontend	Existing React B2C shop (API-only backend)
Estimate	About 420 hours for v1 (Phases 1–3), including B2B APIs and buffer
Status	Draft plan for implementation

This document describes a new B2C backend for Mazing. It is a separate Laravel application with its own MySQL database. It follows the same engineering family as B2B (Laravel, queues, Sanctum, Zoho Payments) but is not a fork of the B2B repository. B2C sells and takes payment. Fulfilment stays on B2B. There is no SalesZing, no Zoho Books, and no challans, split orders, POs, or manager GST signup on B2C.

1. Decisions

Topic	Decision
Framework	Latest Laravel (11/12), not Laravel 8
B2B-only operations	Not on B2C: challans, split orders, POs, manager GST signup, packing, e-way, statements
M-Coins	Yes — customer points wallet on B2C (not B2B salesman incentives)
Catalog	Daily pull copy from B2B; B2C may override locally
Accounting	None on B2C for now (no SalesZing, no Zoho Books)
Payments	Zoho Payments (payment link + webhook)

After order	Push paid order to B2B; B2B processes fulfilment
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2. Architecture

React (B2C) talks only to the B2C Laravel API. That API talks to Zoho Payments for checkout, pulls products from B2B on a schedule, and pushes paid orders back to B2B for fulfilment.

React (B2C shop) → B2C Laravel API → Zoho Payments

B2C Laravel API → daily product pull → B2B Laravel (read-only product feed)

B2C paid orders → order ingest API → B2B (warehouse / invoice fulfilment)

Two databases. B2C never writes B2B tables directly.

B2B must add two small, clean APIs. Do not hang these off OrderController or PurchaseOrderController:

1. Product feed — published items, warehouse stock, prices, images. Business key: part_no.
2. Order ingest — create a B2B order from a paid B2C order (customer snapshot, line items, warehouse, payment reference).

3. What each system owns

B2C owns

- Shoppers, addresses, cart, wishlist, checkout
- B2C orders and payment state
- M-Coin ledger (customer loyalty wallet)
- Offers, banners, homepage merchandising
- Local product overrides (title, sell price, images, description, visible_on_b2c)

B2B owns

- Product master and real warehouse stock
- Fulfilment (existing order → warehouse → invoice flow)
- Zoho Books (unchanged)

Shared contract

- part_no as the product key
- B2C order code (for example B2C-...) stored on the B2B order as source / reference

4. Product sync

A nightly job on B2C copies catalog from B2B:

3. Pull B2B feed (updated_since or full snapshot).
4. Upsert local products and stocks keyed by part_no.

5. Do not overwrite B2C override columns.
6. New SKUs: either unpublished until merchandised, or auto-publish. Pick one rule and keep it.

Local overlay table (product_overrides): sell price, compare-at price, images, description, badges, is_published.

Stock: a daily copy is not enough at checkout. At pay time, check local synced quantity, decrement, and reject if insufficient. A live B2B stock check can be added later; it is not required for v1.

5. Order and payment flow

7. Customer checks out on React (OTP user, address, cart).
8. B2C creates an order in pending_payment.
9. B2C creates a Zoho Payment link (same idea as B2B generatePaymentUrl). Use the B2C order number instead of a B2B invoice or party_code.
10. Webhook marks the order paid, decrements stock, and awards M-Coins.
11. Queue a handoff to B2B. Retry. Make it idempotent on B2C order id.
12. B2B creates its order and processes internally. B2C stores b2b_order_id / status if B2B can callback later.

No credit-days check, no SalesZing due check, no challan. Version 1 is prepaid Zoho only. COD is optional later.

If handoff fails: the order stays paid on B2C. Alert admin and retry. Do not double-charge.

6. M-Coins (B2C wallet)

This is customer loyalty, not B2B MCoin / ManagerMCoin (those are salesman bonuses on invoices and overdue). Do not copy CalculateManagerMCoin.

- m_coin_wallets — running balance per shopper
- m_coin_transactions — earn / redeem / expire, linked to order
- Earn on paid order (config: percent of net, or flat)
- Redeem at checkout (cap, for example max percent of cart)
- Admin: adjust, expiry, history

7. B2C API for React (/api/v1/...)

Public

Home, categories/brands, search, product detail, offers, banners.

Authenticated (Sanctum + OTP)

Profile, addresses, wishlist, cart, checkout, orders, M-Coin balance and history.

Payments

Create Zoho payment link. Webhook endpoint that verifies the Zoho payload.

Staff

Dashboard counts, B2C orders, product overrides, offers, M-Coin adjustments.

Keep controllers small. Put Zoho, product sync, and B2B handoff in services.

8. B2C admin (minimal)

- Orders: status, payment, B2B handoff state
- Catalog overrides and publish flag
- Offers / coupons
- M-Coin rules and manual credit
- Users

No purchase orders, packing, Zoho Books, or salesman dashboard.

9. Core data model

users, addresses, products (synced), product_overrides, product_stocks, carts, wishlists, orders, order_items, payments (Zoho), m_coin_wallets, m_coin_transactions, offers / coupons, banners.

Do not create: challan, sub_order, packing_*, purchase_*, Zoho Books tokens, or required party_code for shoppers. Shoppers are not B2B dealers.

Handoff customer model (operations to choose): a new B2B user per shopper, or one “Mazing Store B2C” account plus a shipping snapshot on the order. No GST signup on B2C.

10. What not to copy from B2B

- Laravel 8 / Active eCommerce CMS addons
- Seller / marketplace routes and unused payment gateways (Paytm, Stripe, and similar)
- Nested api/*.php SalesZing scripts
- God controllers and hardcoded user IDs
- SalesZing, Zoho Books, statements, 41 Manager, import PO

Reuse as reference only

- Sanctum + OTP login pattern
- Product field names (part_no, HSN, tax, images)
- Zoho Payments link + webhook pattern
- Queues and jobs

11. Delivery phases

Phase	Scope
1 — Shop	Auth, catalog (sync + overrides), product detail, cart, wishlist, checkout, Zoho Pay, order history, basic admin orders

2 — B2B wiring	Daily product job, paid-order ingest, handoff retries, stock decrement, publish flags
3 — M-Coins + merchandising	Earn / redeem, offers, banners, reward SKUs if needed
4 — Later (not now)	Live stock API, order-status callback from B2B, Zoho Books, COD

12. Estimated timeline (hours)

Estimates assume one experienced Laravel developer, the React shop already built (backend API work only), a greenfield Laravel 11/12 app, and Zoho Payments following the existing B2B payment-link pattern. Hours include implementation, basic automated tests, and wiring to B2B. They do not include React UI changes, content/merchandising, or production infra beyond a standard deploy.

Phase totals

Phase	Low (h)	Likely (h)	High (h)	Calendar (1 backend)
1 — Shop (B2C APIs + admin + Zoho Pay)	160	180	210	4.5–5.5 weeks
2 — B2B product feed + order ingest + sync	80	100	130	2.5–3.5 weeks
3 — M-Coins, offers, banners	60	80	100	1.5–2.5 weeks
QA, hardening, buffer (≈15%)	45	55	70	1–1.5 weeks
Total v1 (Phases 1–3)	345	415	510	10–13 weeks

Likely v1 total: about 415 hours (~10–11 weeks at 40 hours/week, one backend engineer).

Breakdown by module

Work	Where	Hours
Project setup: Laravel 11/12, Sanctum, queues, structure, env	B2C	16
Auth: OTP login/signup, profile, addresses	B2C	28
Catalog models, search/list/PDP APIs, categories/brands	B2C	36
Product overrides + publish flag + admin catalog	B2C	20
Cart + wishlist	B2C	20
Checkout + order create + order history APIs	B2C	28
Zoho Payments: link, reuse, webhook, paid/failed, retries	B2C	32
Staff admin: orders, users, dashboard counts	B2C	24
B2B product feed API (read-only, part_no, stock, images)	B2B	20
B2C daily/hourly sync job + merge without clobbering overrides	B2C	20
B2B order ingest API (idempotent, shipping snapshot)	B2B	36
Handoff queue, retries, alerts, store b2b_order_id	B2C	16
Stock decrement at payment + oversell guard	B2C	10
M-Coin wallet, earn, redeem at checkout, expiry	B2C	32
M-Coin admin adjust + history	B2C	12

Offers / coupons APIs + checkout apply	B2C	24
Banners / home merchandising APIs	B2C	12
QA, bugfix, deploy hardening (buffer)	Both	55
Total v1		441

The module table sums to about 441 hours (mid-range). Use 415 as the planning number and 510 as the ceiling if B2B ingest is slower than expected (that controller layer is large and easy to under-estimate).

Phase 4 (not in v1)

Item	Hours (later)
Live B2B stock check at checkout	16–24
Order-status callback from B2B to B2C	16–24
COD	12–20
Zoho Books on B2C (if ever required)	40–80

Assumptions and risks that change hours

- React already exists and only needs stable JSON contracts. Extra frontend work is not in this number.
- B2B order ingest is a new thin API, not a rewrite of OrderController. If ingest must mimic full B2B split/challan rules, add 40–80 hours.
- One fulfilment warehouse for v1. Multi-warehouse routing adds ~16 hours.
- M-Coin rules stay simple (earn % + redeem cap). Complex tiers/expiry campaigns add ~16–24 hours.
- Zoho Payments credentials and webhook access are available on day one. Sandbox delays sit outside the estimate.
- Hours are development hours, not elapsed calendar if the developer is split across B2B support.

13. Risks to settle before build

13. Handoff customer: new B2B user per shopper vs one B2C dummy customer.
14. Price: B2B MRP vs B2C sell price (overrides).
15. Which warehouse B2C sells from (one DC vs nearest).
16. Sync window vs oversell if stock is not reserved at checkout.
17. M-Coin earn rate, redeem cap, and expiry — product needs numbers.

14. Summary

Build a latest-Laravel B2C shop on its own server. Keep a local catalog copy, synced daily from B2B, with local overrides. Take payment with Zoho Payments. Run an M-Coin customer wallet. Push paid orders to B2B for fulfilment. Do not put an accounting stack or B2B operations features on this server.

Plan on about 415 hours for v1 (roughly 10–11 weeks for one backend engineer), with a ceiling near 510 hours if B2B order ingest is harder than a thin API.

This plan is based on the current B2B Laravel codebase and client answers on framework, M-Coins, product sync, accounting, and Zoho Books. Hour estimates are for backend development unless noted.
